

IMO 2026 — Problem 6

Solution by xAI Grok 4.5

2026-07-21

Source: `results/grok-4.5/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

partial

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Approaches tried

- Computational exploration of many starting values ($a_1 = 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 14, 15, 21, 25, 27, 30, 33, 35, 45, 49, 75$)
 - When a_1 is a prime power p^k , the sequence becomes all multiples of p from a_1 onwards: $T=1, L=p$.
 - When a_1 is even (or shares factor 2 early), often collapses to all even numbers: $T=1, L=2$.
 - When $a_1=15=3 \cdot 5$: *period* $T=8, L=30=2 \cdot 3 \cdot 5$, residues $\{0, 6, 10, 12, 15, 18, 20, 24\} \pmod{30}$.
 - When $a_1=35=5 \cdot 7$: *period* $T=34, L=210=2 \cdot 3 \cdot 5 \cdot 7$.

- Pattern: finitely many “essential” primes seem to determine an eventual modulus $L =$ product of them, and the set of admissible residues is closed under $+L$.
- Key observation: a number m is a candidate for future terms iff $\gcd(m, a_i) > 1$ for all previous a_i . Equivalently, for every previous a_i , m shares a prime factor with a_i .
- Let P_n be the set of all primes dividing at least one of $a_1, \dots, a_n$. Then the condition is stronger than “ m shares a prime with the product”; m must hit each a_i individually.
- New primes keep appearing (infinitely many, by computational evidence), but they appear as “optional” factors on numbers already forced by the finite set of early primes.
- Conjecture: there is a finite set of primes S (the “core” primes that appear early and cover the sequence) such that $L = \text{prod}(S)$ (or lcm of early terms), and eventually the sequence consists exactly of all integers $>$ some point that are divisible by at least one prime from each of certain covering sets — actually more structured: the admissible set becomes a finite union of arithmetic progressions with difference L , and the sequence enumerates them in order.

Current best

Empirically the sequence is always eventually periodic in the strong sense $a_{n+T} = a_n + L$. The core idea is that only finitely many primes are “necessary” for the covering condition; later primes appear but don’t restrict the admissible residue classes mod L further. Need a rigorous proof that (1) only finitely many primes can ever be forced as new restrictions, or more carefully: the set A of all positive integers m that share a factor with every a_i (i.e., admissible w.r.t. the whole infinite sequence) is a finite union of APs, and the sequence is exactly the increasing enumeration of A ($a_1 +$) or similar — but wait, that’s circular because A depends on the sequence. Better: track the admissible set after n steps, show it stabilizes in terms of residue classes after finitely many steps.

Full proof

(Not yet complete.)